

ECON 527: Problem Set 1 Answer Key

1 Discretizing an AR(1)

In this problem, we will examine the accuracy of various methods for approximating a highly persistent AR(1) process

$$\begin{aligned}x_t &= \rho x_{t-1} + \varepsilon_t \\ \varepsilon_t &\sim N(0, \sigma^2)\end{aligned}\tag{1}$$

where $\rho = 0.99$ and $\sigma = 0.2$.

1. Use Tauchen's method with 9 gridpoints and 3 standard deviations to discretize (1).
2. Simulate 1,000 histories, each 1,000 periods long, of the resulting Markov chain. Draw the initial states from the invariant distribution. Compute each of the following moments from the simulation, and report the ratio between the simulated and true moment:
 - (a) First-order autocorrelation: $\text{cor}(x_t, x_{t-1})$
 - i. Simulated: 0.9986
 - ii. True: $\rho = 0.99$
 - iii. Ratio: $0.9986/0.99 = 1.0087$
 - (b) Unconditional variance: $\text{var}(x_t)$
 - i. Simulated: 3.1947
 - ii. True: $\sigma^2/(1 - \rho^2) = 2.0101$
 - iii. Ratio: $3.1947/2.0101 = 1.5893$
 - (c) First-order autocovariance: $\text{cov}(x_t, x_{t-1})$
 - i. Simulated: 3.1903
 - ii. True: $\text{var}(x_t)\rho = 1.9899$
 - iii. Ratio: $3.1903/1.9899 = 1.6032$
 - (d) Conditional variance: $\text{var}(x_t|x_{t-1})$
 - i. Simulated: 0.0089

- ii. True: $\sigma^2 = 0.0400$
 - iii. Ratio: $0.0089/0.0400 = 0.2236$
3. Repeat (1) and (2) using Rouwenhorst's method. Which method is more accurate in this case?
- (a) First-order autocorrelation: $\text{cor}(x_t, x_{t-1})$
 - i. Simulated: 0.9900
 - ii. Ratio: $0.9900/0.99 = 1.0000$
 - (b) Unconditional variance: $\text{var}(x_t)$
 - i. Simulated: 2.0041
 - ii. Ratio: $2.0041/2.0101 = 0.9971$
 - (c) First-order autocovariance: $\text{cov}(x_t, x_{t-1})$
 - i. Simulated: 1.9841
 - ii. Ratio: $1.9841/1.9899 = 0.9971$
 - (d) Conditional variance: $\text{var}(x_t|x_{t-1})$
 - i. Simulated: 0.0398
 - ii. Ratio: $0.0398/0.0400 = 0.9955$

Rouwenhorst's method is more accurate in this case. This is consistent with what we discussed in class and was argued by Kopecky and Suen (2010): Rouwenhorst's method is more accurate than Tauchen's method for highly persistent processes.

2 Interpolation

In this problem, we will practice interpolation using the CRRA utility function

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

with curvature parameter $\gamma = 4$.¹

1. Make an evenly spaced grid of consumption values c_i with $N = 100$ grid-points over the interval $[0.1, 30]$. Use this grid to create a natural cubic spline interpolant $g(c)$ of the true function $u(c)$.
2. Evaluate the accuracy of this interpolating function as follows:

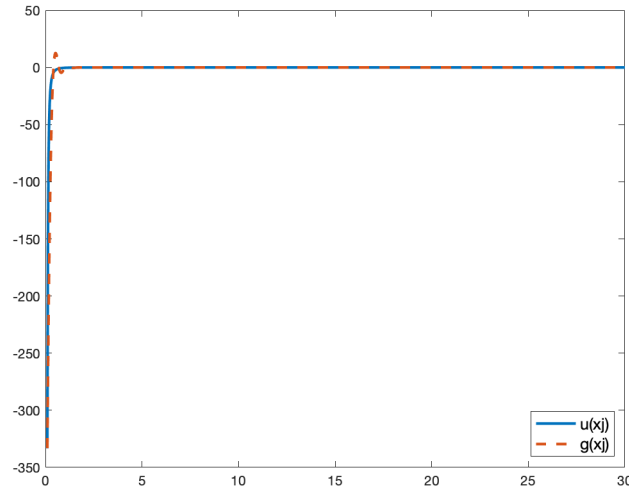
¹This value is perhaps on the high side, but well within the range of values commonly used in macroeconomic studies.

- (a) Make a fine, evenly spaced grid x_j with 1000 gridpoints over the interval $[0.1, 30]$. Compute the error

$$f(x_j) = |u(x_j) - g(x_j)|$$

at each gridpoint. Plot (in the same graph) $u(x_j)$ and $g(x_j)$. Report the maximum error $\max_j f(x_j)$.

- i. Graph:



- ii. Max error: 141.6

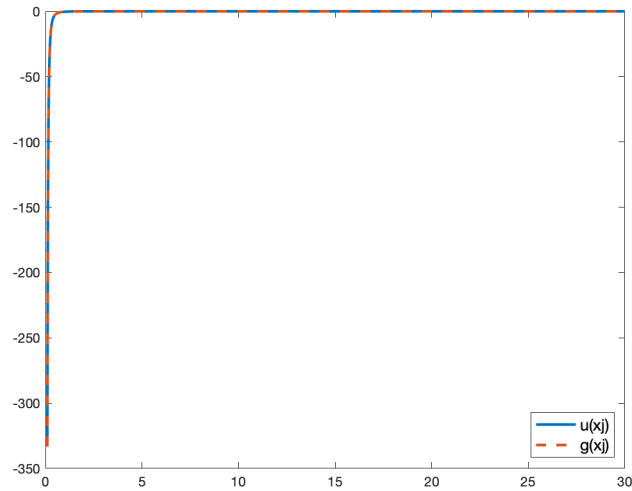
- (b) We know that $u(c)$ is monotonically increasing, strictly concave, and less than 0 over the interval $[0.1, 30]$. Is this true of $g(x_j)$? In other words, is it the case that:

$$\begin{aligned} g(x_{j+1}) - g(x_j) &> 0 \\ \frac{g(x_{j+2}) - g(x_{j+1})}{x_{j+2} - x_{j+1}} &< \frac{g(x_{j+1}) - g(x_j)}{x_{j+1} - x_j} \\ g(x_j) &< 0 \end{aligned}$$

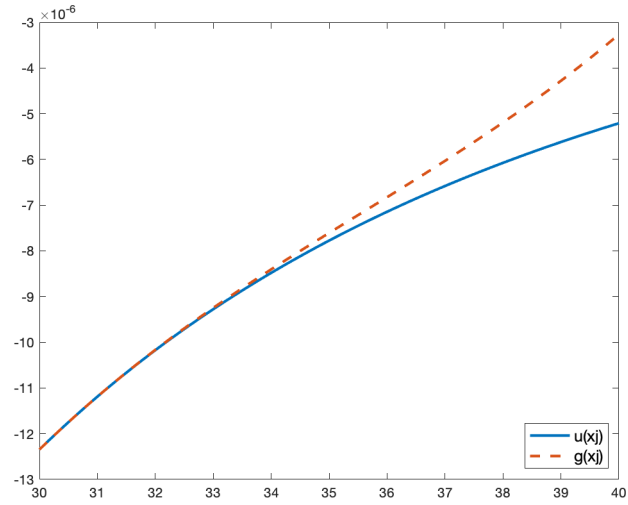
- i. Monotonically increasing: no
 ii. Strictly concave: no
 iii. Less than 0: no
- (c) Does the interpolating function g provide a good approximation of the true function u ?
- i. No. It is inaccurate and does not preserve important function properties.

3. Repeat (1) and (2) but instead of using an evenly spaced grid to create the interpolating function, use a polynomial grid with curvature parameter 3 (use the same number of gridpoints and grid boundaries as before).

(a) Graph:



- (b) Max error: 0.0050
 - (c) Monotonically increasing: yes
 - (d) Strictly concave: yes
 - (e) Less than 0: yes
 - (f) Good approximation? Yes - small error and preserves important function properties
4. Now let's test the accuracy of the interpolating function created for (3) when used for extrapolation.
 - (a) Plot (in the same graph) $u(x_j)$ and $g(x_j)$ on an evenly spaced x_j grid with 100 nodes over the interval $[30, 40]$. What is the maximum error over this range? Are monotonicity, concavity, and the fact that $u(c) < 0$ preserved?
 - i. Graph:

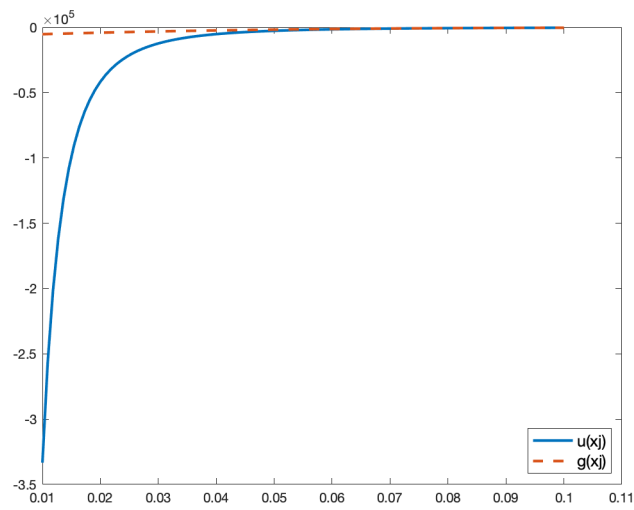


ii. Max error: 1.94×10^{-6}

iii. The extrapolation preserves all of the properties except strict concavity.

(b) Repeat (a) for the interval $[0.01, 0.1]$

i. Graph:



ii. Max error: 3.28×10^5

iii. The extrapolation preserves all of the properties.

(c) Is extrapolation over the interval $[0.01, 0.1]$ more or less accurate than extrapolation over the interval $[30, 40]$? Why?

- i. Less accurate. The true function is highly curved over this range, whereas it is nearly linear over $[30, 40]$. Even when the true function is approximately linear, extrapolation does not preserve concavity. This is an illustration of the dangers of extrapolation.